E-COMMERCE SALES DATA ANALYSIS

Exploratory Data Analysis (EDA) Project

Dataset: Sample Superstore
Analysis Period: 2014 – 2017
Tools: Python, Pandas, NumPy, Matplotlib, Seaborn

Final Project Report

1. Introduction

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset using statistical summaries and visualizations. This project analyzes an E-Commerce Sample Superstore dataset to identify sales trends, profitability patterns, and business insights.

2. Project Objectives

• Understand the structure of the sales dataset.

• Check and handle missing values and duplicate records.

• Analyze sales and profit across categories and regions.

• Study monthly and yearly sales trends.

• Understand the relationship between discounts and profit.

• Generate visualizations and meaningful business insights.

3. Dataset Description

Metric	Value
Total Records	9,994
Original Columns	21
Columns After Feature Creation	23
Total Customers	793
Total Products	1,862
Categories	3
Total Sales	2,297,200.86
Total Profit	286,397.02
Analysis Period	2014–2017

4. Data Cleaning and Preprocessing

• Loaded the Sample Superstore CSV dataset using Pandas.

• Converted Order Date and Ship Date into datetime format.

• Checked all columns for missing values; none were found.

• Checked duplicate records; zero duplicates were found.

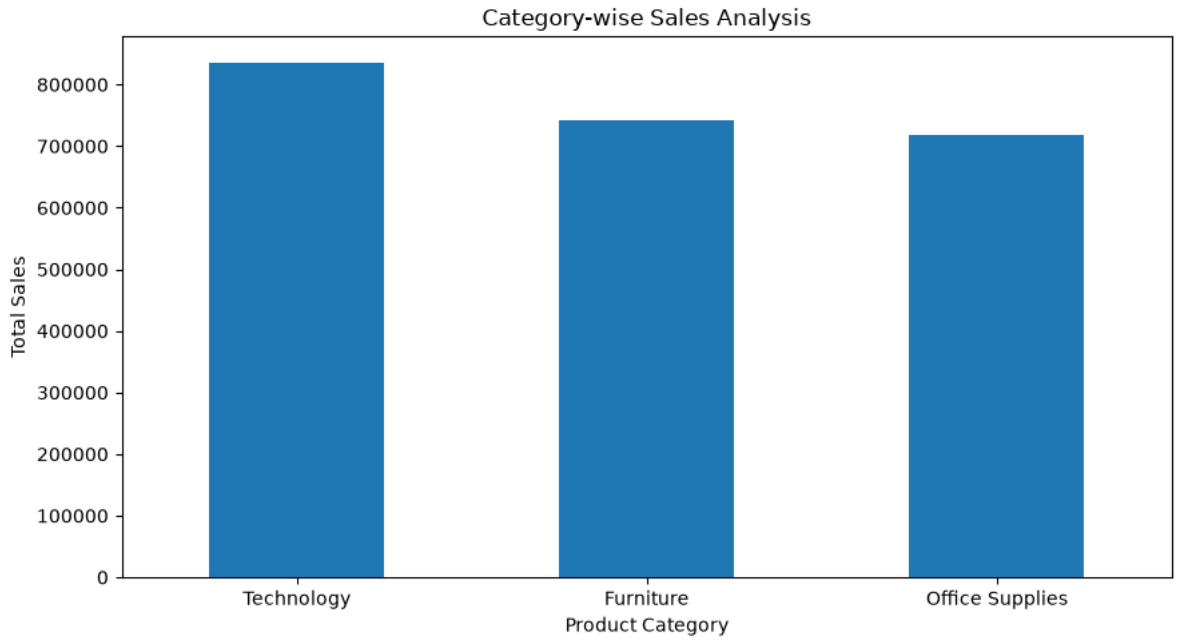
• Created Year and Month columns from Order Date for time-based analysis.

• Dataset contained 9,994 records after preprocessing.

5. Exploratory Data Analysis and Visualizations

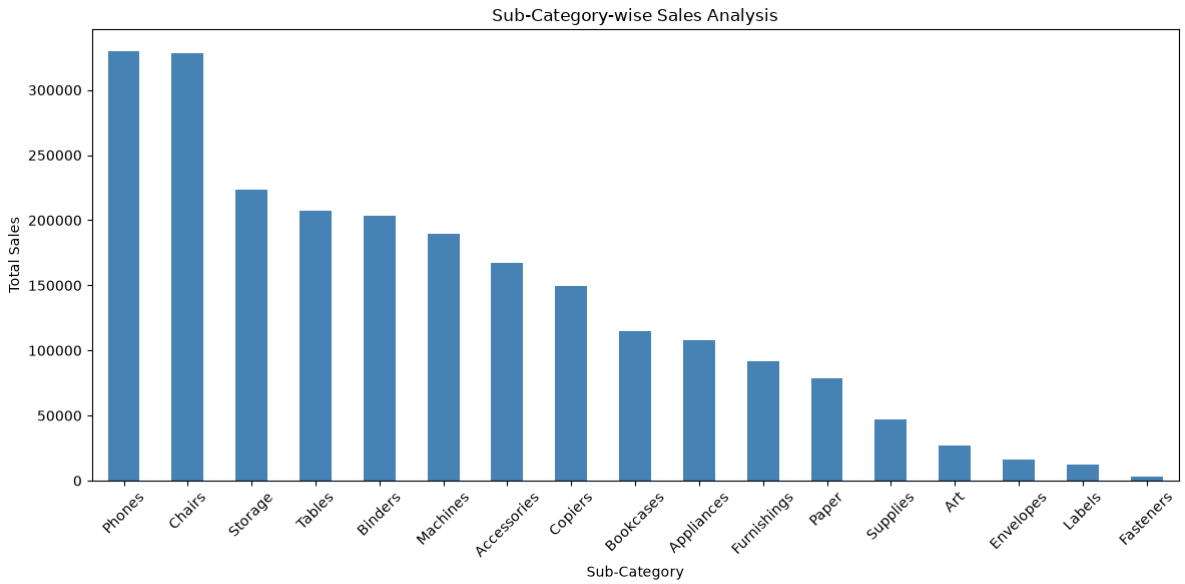
Category-wise Sales

Technology recorded the highest sales among the three product categories.



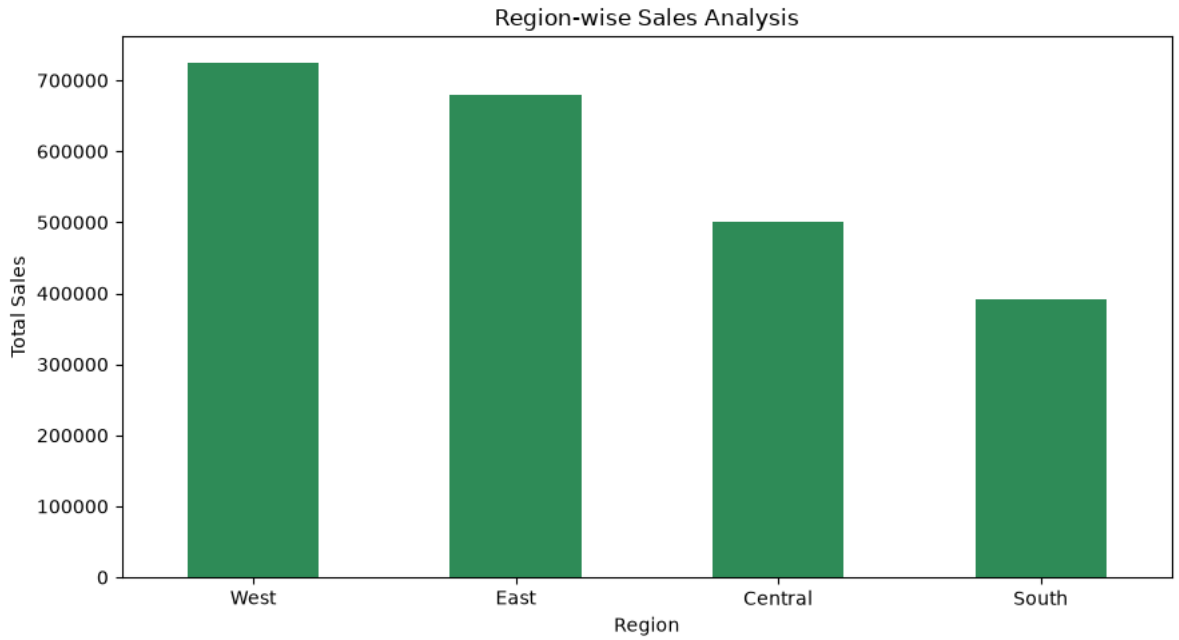
Sub-Category-wise Sales

Phones recorded the highest sales, while Fasteners recorded the lowest.



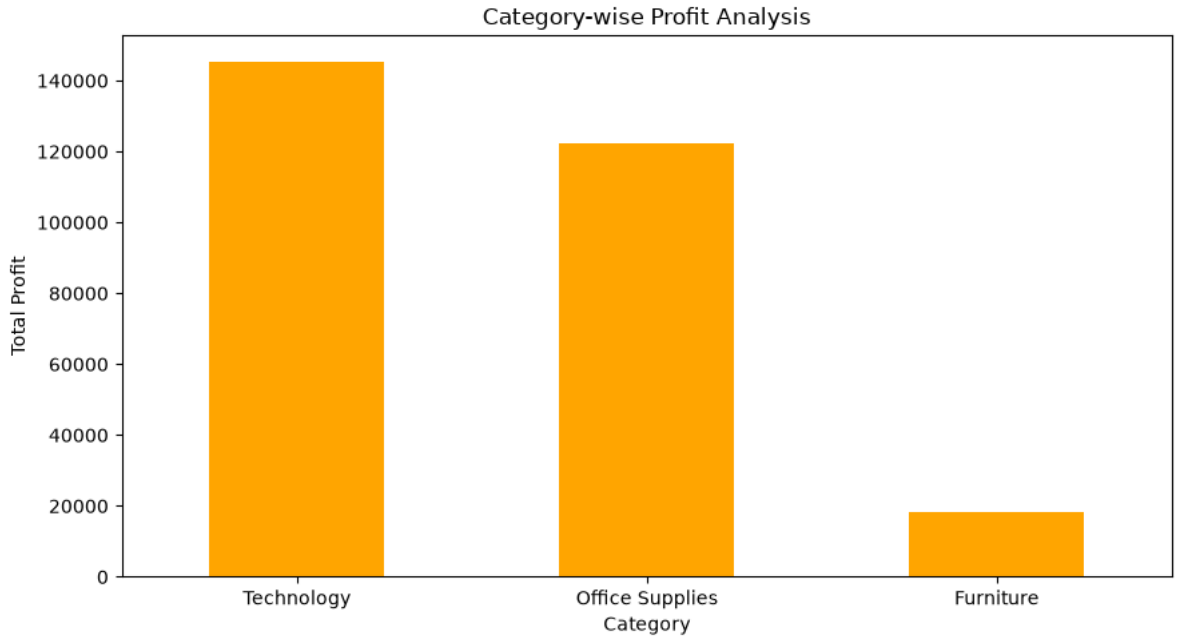
Region-wise Sales

The West region generated the highest sales, while the South generated the lowest.



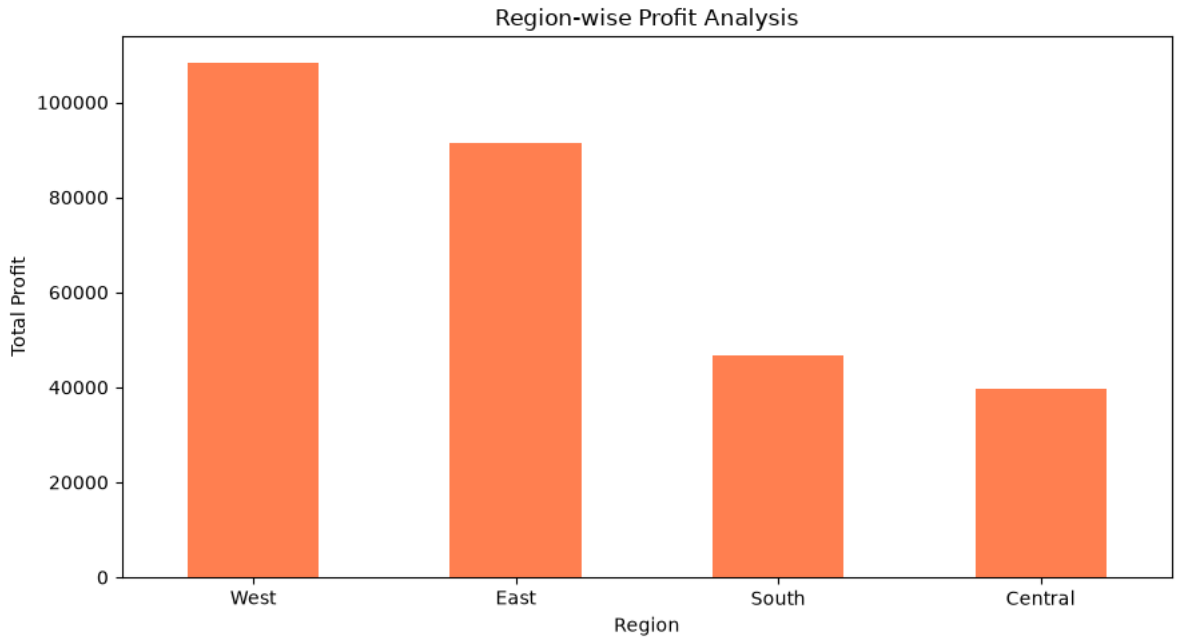
Category-wise Profit

Technology generated the highest profit. Furniture generated substantially lower profit.



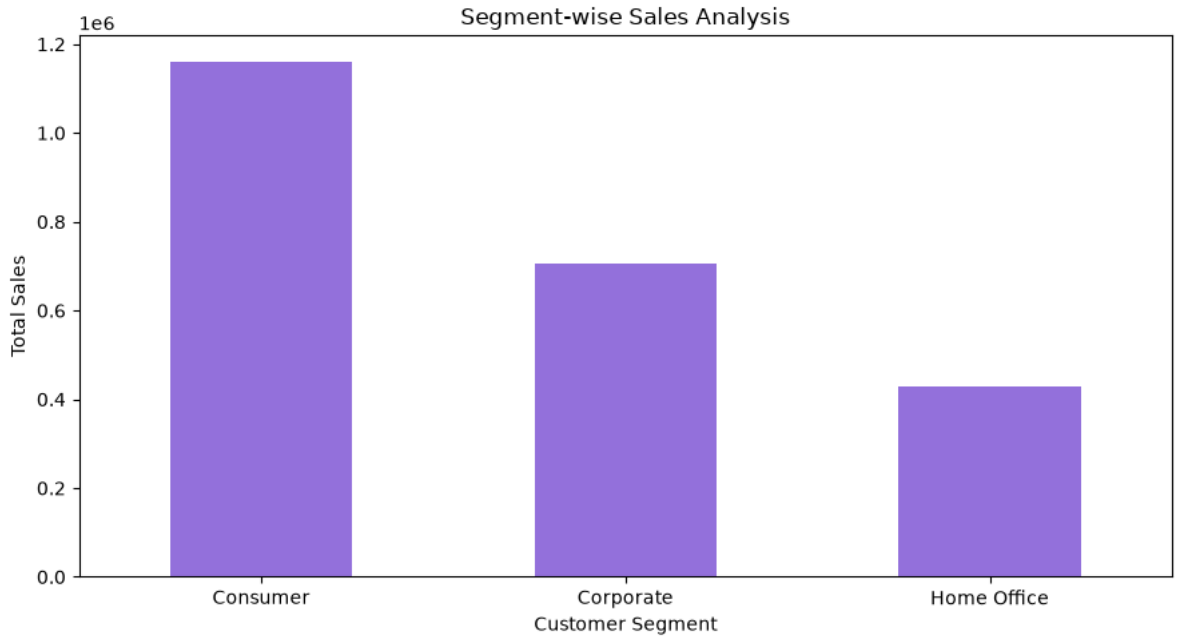
Region-wise Profit

The West region generated the highest profit, while Central recorded the lowest.



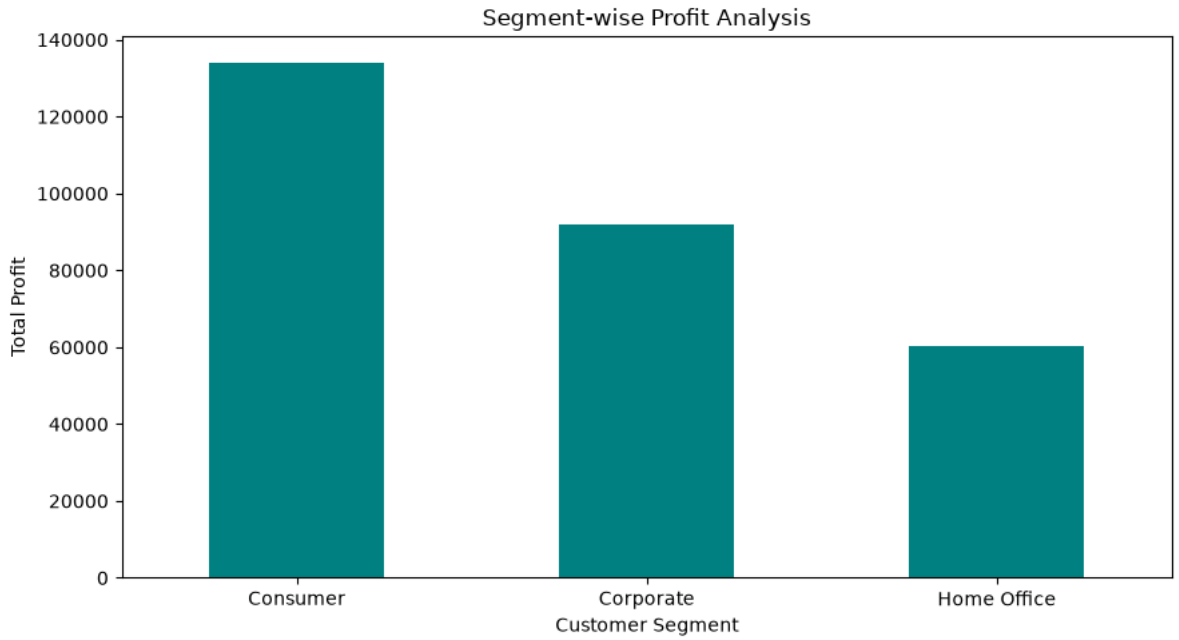
Segment-wise Sales

The Consumer segment generated the highest sales.



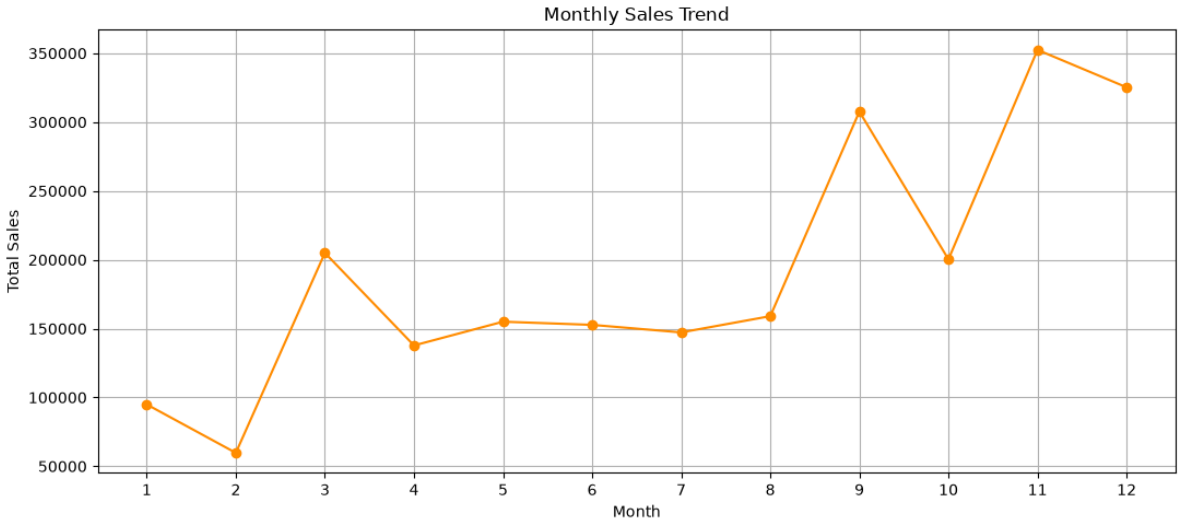
Segment-wise Profit

The Consumer segment generated the highest profit.



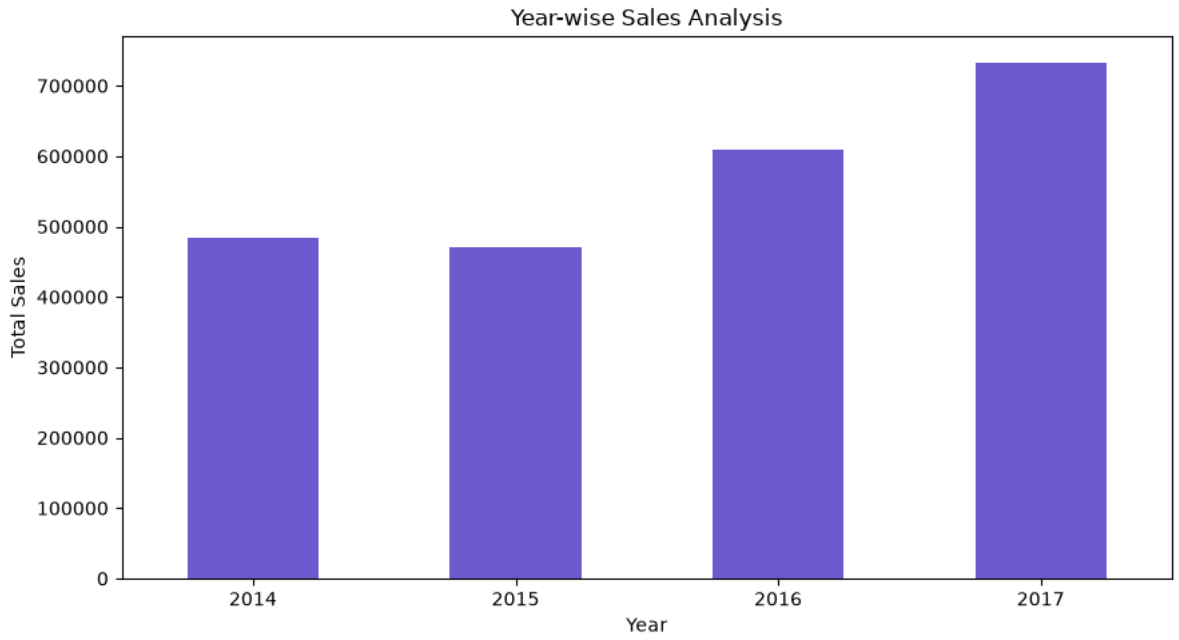
Monthly Sales Trend

November recorded the highest monthly sales, while February recorded the lowest.



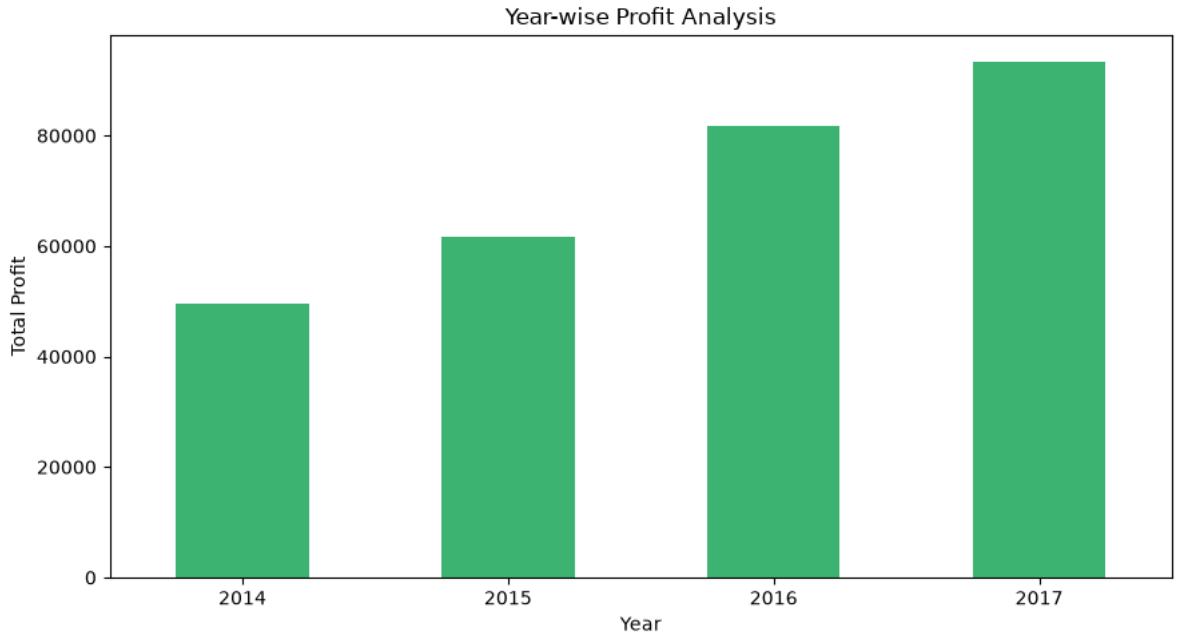
Year-wise Sales

2017 recorded the highest annual sales during the analysis period.



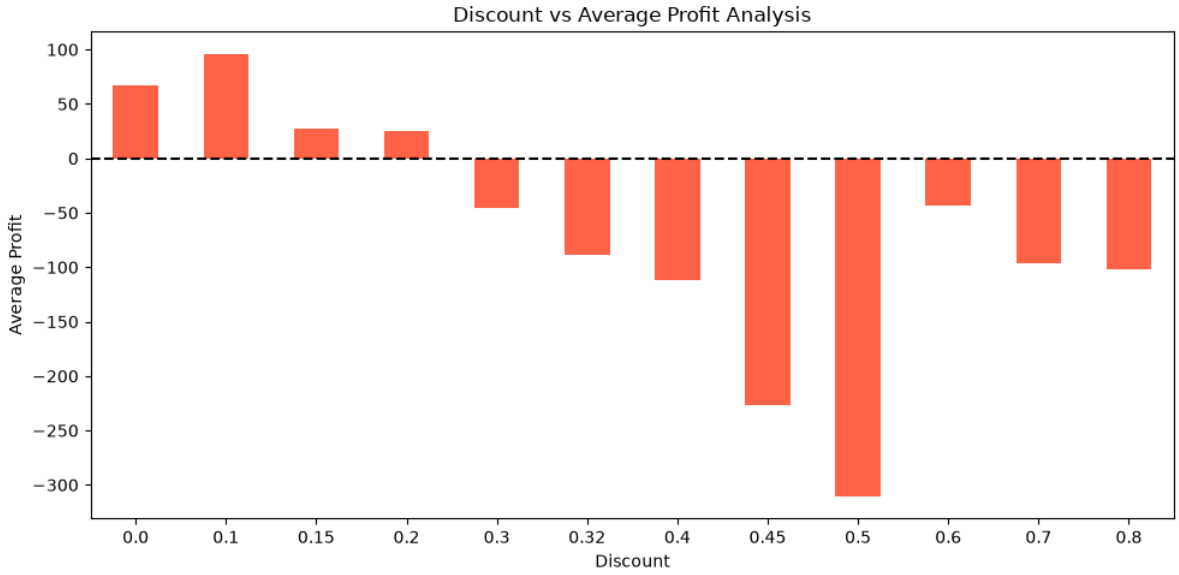
Year-wise Profit

Annual profit increased from 2014 through 2017.



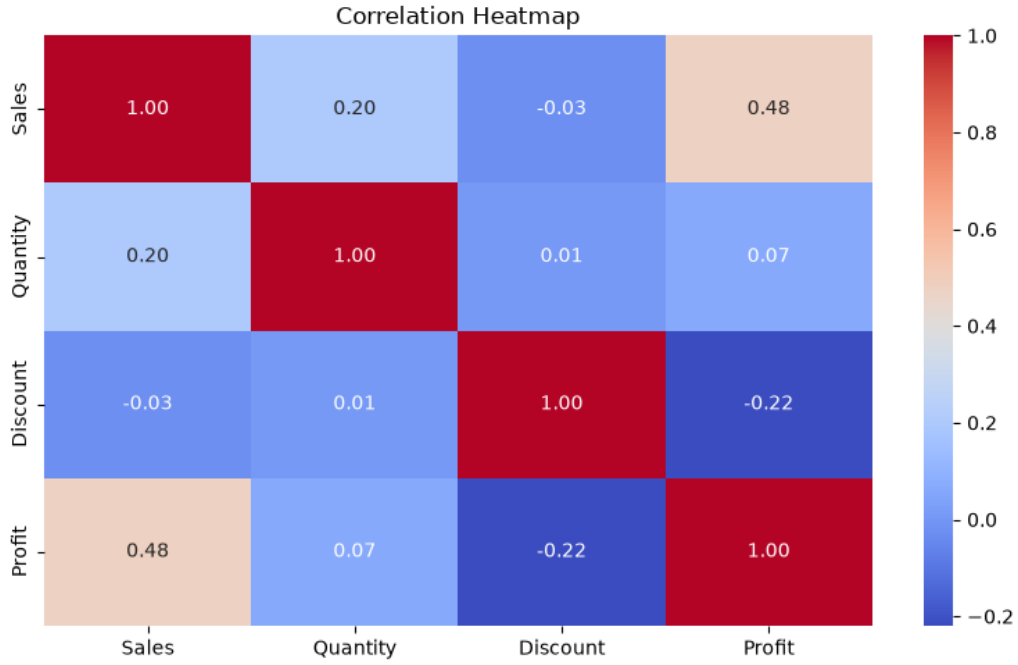
Discount vs Average Profit

A 10% discount had the highest average profit. A 50% discount had the lowest average profit.



Correlation Heatmap

The heatmap displays correlations among Sales, Quantity, Discount, and Profit.



6. Key Business Findings

• Technology is the highest sales and profit generating category.

• West is the leading region in both sales and profit.

• Consumer is the highest sales and profit generating segment.

• 2017 recorded the highest annual sales.

• A 10% discount had the highest average profit.

• A 50% discount had the lowest average profit.

7. Conclusion

The EDA project provided useful insights into the Sample Superstore dataset. Technology and the West region performed strongly in terms of sales and profit. The Consumer segment contributed the highest sales and profit, and 2017 was the strongest year in the dataset. The discount analysis showed that higher discounts can be associated with lower average profit at several discount levels. Overall, this analysis demonstrates how Python-based data analysis and visualization can help identify business trends and support better decisions.